Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete tiny-image development and freeze record.

Trace fixed/trained components,select the saved state and distinguish development from teacher-released final evidence.

Keep one habit: W1–W3/W5 → selected new model → actual project freeze/evaluation ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

  This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

- Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

  Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

- For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

  Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

Pooling and spatial information

> **My remembered idea, example and correction**

Image tensors and channels

My remembered idea, example and correction

Initialisation and reproducibility

My remembered idea, example and correction

Lesson reference for workbook practice

Split and ID	Image rows	Class	Difference d
Train A	[1, 0] and [0, 0]	1	1
Train B	[1, 0] and [1, 0]	1	2
Train C	[0, 1] and [0, 0]	0	-1
Train D	[0, 1] and [0, 1]	0	-2
Validation E	[0, 0] and [1, 0]	1	1
Validation F	[0, 0] and [0, 1]	0	-1

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete pixel inputs for W1

Split and ID	Image rows	Class	Difference d
Train A	[1, 0] and [0, 0]	1	1
Train B	[1, 0] and [1, 0]	1	2
Train C	[0, 1] and [0, 0]	0	-1
Train D	[0, 1] and [0, 1]	0	-2
Validation E	[0, 0] and [1, 0]	1	1
Validation F	[0, 0] and [0, 1]	0	-1

Complete task and evidence

- Main valid 2×2 binary images use frozen K[[1, -1], [1, -1]] to compute left-column sum minus right-column sum d. Hidden values are [ReLU(d), ReLU(-d)]; score a(h1-h2)=ad. Only a is trained; output coefficients [a, -a] are tied. Training A/B/C/D have d1, 2, -1, -2 and signed targets 1, 1, -1, -1. Validation E/F have d1, -1 and signed targets 1, -1. Start a0, rate.1, two full-data updates; gradient 5a-3.

  Saved weights 0, .3, .45 have training MSE 1, .325, .15625 and validation MSE 1, .49, .3025. Minimum validation MSE selects update 2, a≈.45. Both positive checkpoints have validation class accuracy 2/2. Training-majority class baseline chooses 0on its 2/2 tie; training 2/4, validation 1/2. Strict score>0 predicts 1, otherwise 0. Public all-zero image gives score 0, class 0.

  Final cases are teacher-held until the actual freeze record; no final score is supplied here. Separate picture uses d2, target 2, rate 1/16 and a0→.5→.75.

- For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main valid 2×2 binary images use frozen K[[1, -1], [1, -1]] to compute left-column sum minus right-column sum d. Hidden values are [ReLU(d), ReLU(-d)]; score a(h1-h2)=ad. Only a is trained; output coefficients [a, -a] are tied. Training A/B/C/D have d1, 2, -1, -2 and signed targets 1, 1, -1, -1. Validation E/F have d1, -1 and signed targets 1, -1. Start a0, rate.1, two full-data updates; gradient 5a-3.

  Saved weights 0, .3, .45 have training MSE 1, .325, .15625 and validation MSE 1, .49, .3025. Minimum validation MSE selects update 2, a≈.45. Both positive checkpoints have validation class accuracy 2/2. Training-majority class baseline chooses 0on its 2/2 tie; training 2/4, validation 1/2. Strict score>0 predicts 1, otherwise 0. Public all-zero image gives score 0, class 0.

  Final cases are teacher-held until the actual freeze record; no final score is supplied here. Separate picture uses d2, target 2, rate 1/16 and a0→.5→.75.

- Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main valid 2×2 binary images use frozen K[[1, -1], [1, -1]] to compute left-column sum minus right-column sum d. Hidden values are [ReLU(d), ReLU(-d)]; score a(h1-h2)=ad. Only a is trained; output coefficients [a, -a] are tied. Training A/B/C/D have d1, 2, -1, -2 and signed targets 1, 1, -1, -1. Validation E/F have d1, -1 and signed targets 1, -1. Start a0, rate.1, two full-data updates; gradient 5a-3.

  Saved weights 0, .3, .45 have training MSE 1, .325, .15625 and validation MSE 1, .49, .3025. Minimum validation MSE selects update 2, a≈.45. Both positive checkpoints have validation class accuracy 2/2. Training-majority class baseline chooses 0on its 2/2 tie; training 2/4, validation 1/2. Strict score>0 predicts 1, otherwise 0. Public all-zero image gives score 0, class 0.

  Final cases are teacher-held until the actual freeze record; no final score is supplied here. Separate picture uses d2, target 2, rate 1/16 and a0→.5→.75.

- Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main valid 2×2 binary images use frozen K[[1, -1], [1, -1]] to compute left-column sum minus right-column sum d. Hidden values are [ReLU(d), ReLU(-d)]; score a(h1-h2)=ad. Only a is trained; output coefficients [a, -a] are tied. Training A/B/C/D have d1, 2, -1, -2 and signed targets 1, 1, -1, -1. Validation E/F have d1, -1 and signed targets 1, -1. Start a0, rate.1, two full-data updates; gradient 5a-3.

  Saved weights 0, .3, .45 have training MSE 1, .325, .15625 and validation MSE 1, .49, .3025. Minimum validation MSE selects update 2, a≈.45. Both positive checkpoints have validation class accuracy 2/2. Training-majority class baseline chooses 0on its 2/2 tie; training 2/4, validation 1/2. Strict score>0 predicts 1, otherwise 0. Public all-zero image gives score 0, class 0.

  Final cases are teacher-held until the actual freeze record; no final score is supplied here. Separate picture uses d2, target 2, rate 1/16 and a0→.5→.75.

- After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

  Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Choose a checkpoint independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Check both updates for **Training** a small image model. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Training** a small image model. Proposed explanation: "All parameters in a trained model must have been learned". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

  Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Train one tiny image score with a fixed budget and a separately reserved check.

- Toy brightness difference d=left−right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/23

Training a small image model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. First update

Gradient 2(2a-2) x 2=-8; new a = 0.5, training loss 1.

2. Second update

Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.

3. Validation score

At reserved d1, target 1: prediction 0.75, squared error 0.0625.

CHANGE ONE THING

Double the two-update budget while keeping the same training rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Train one tiny image score with a fixed budget and a separately reserved check.
- Which labelled step matches this detail? At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.
- 1. Validation score
- 2. First update
- 3. Second update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Validation score At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Changed-case question and optional hint

- Changed-case practice: Double the two-update budget while keeping the same **training** rule. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Repair and retain the record

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
> | | |
> | | |
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

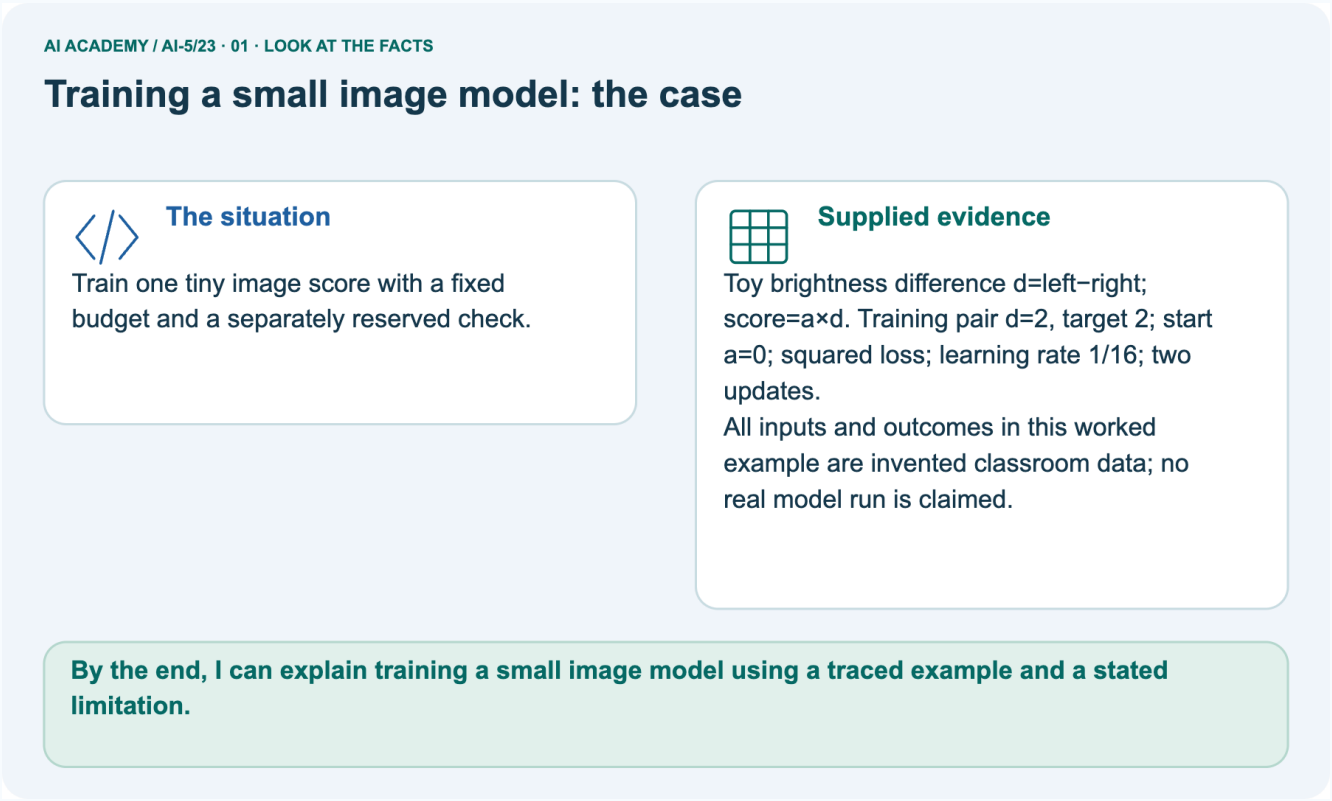
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Training a small image model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

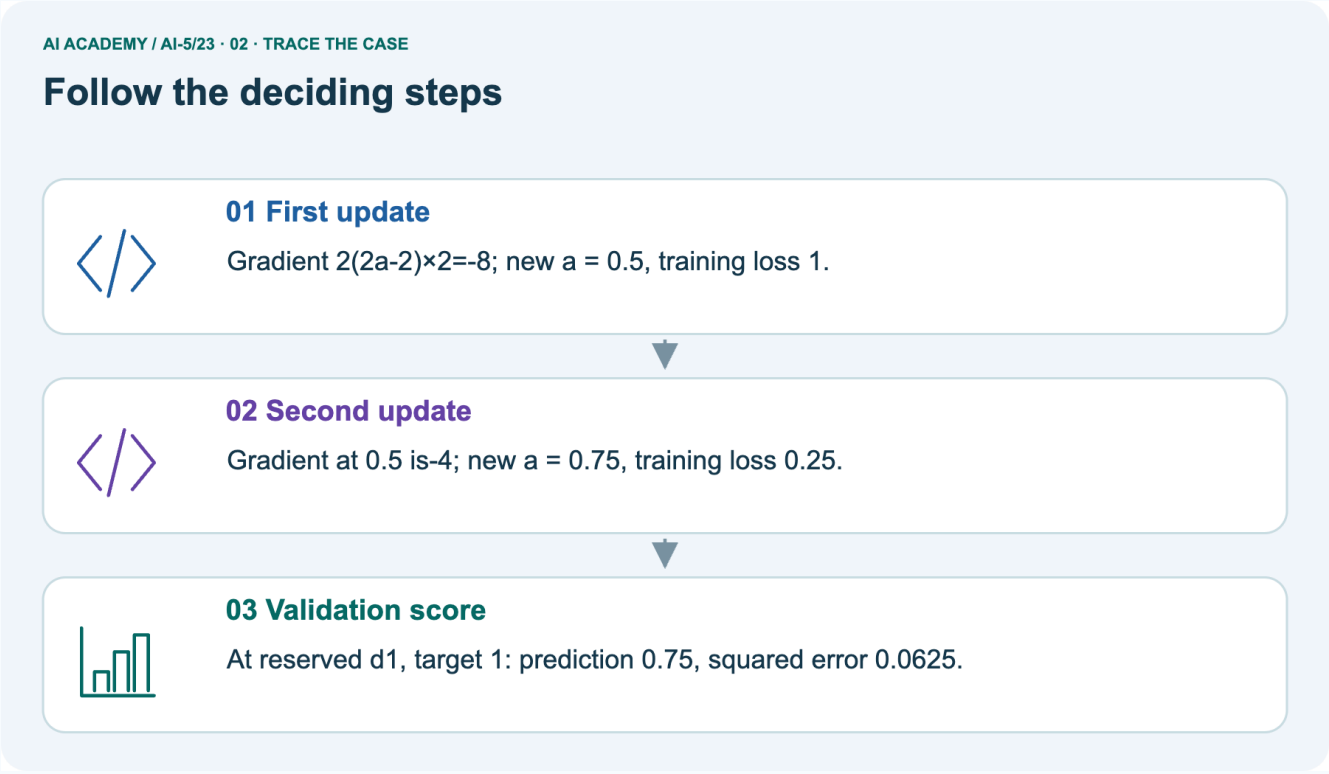
Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

Training pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Compare the saved checkpoints

AI ACADEMY / AI-5/23 · 03 · INSPECT THE LESSON MODEL

Compare the saved checkpoints

Update and weight	Training MSE	Validation MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

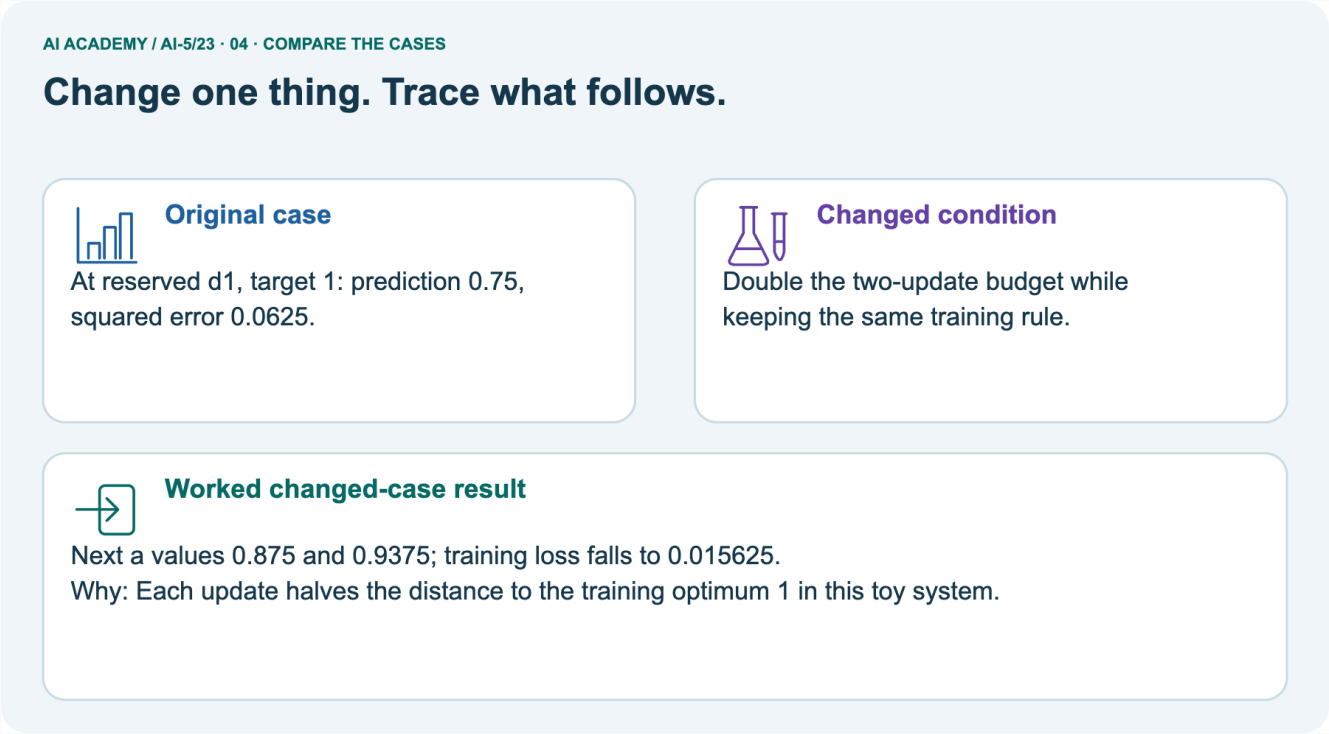
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

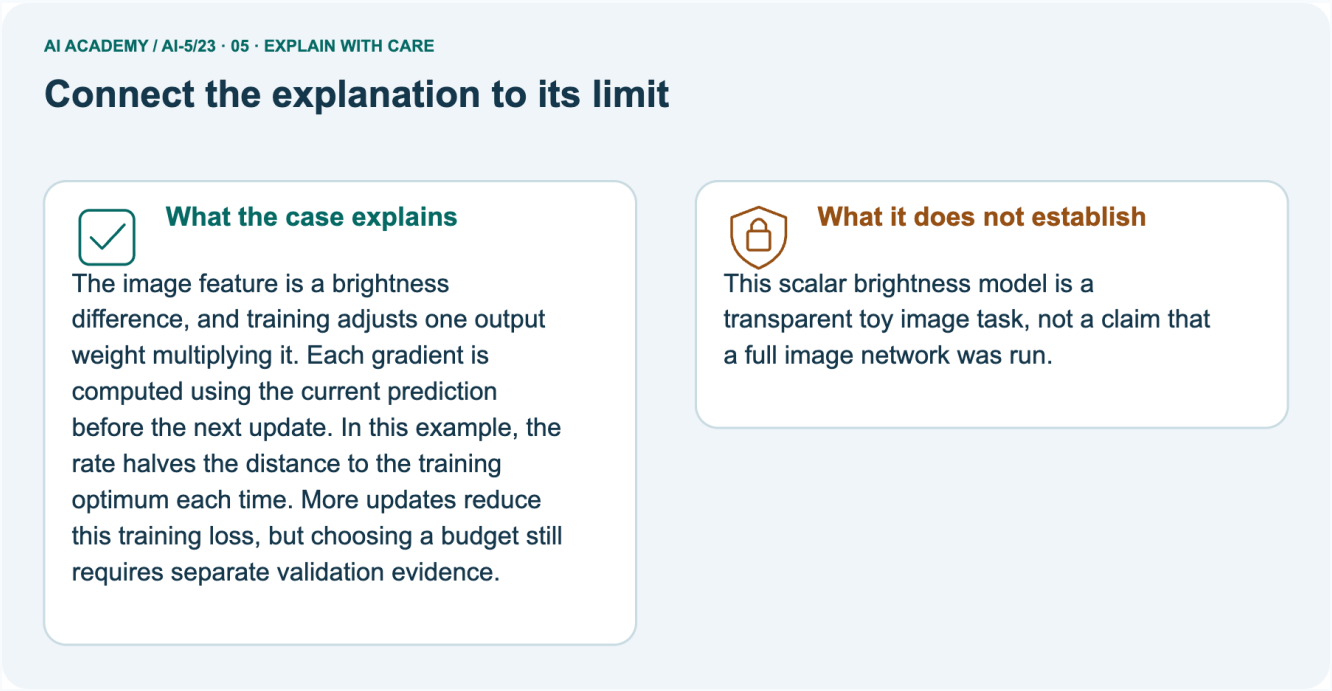
Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

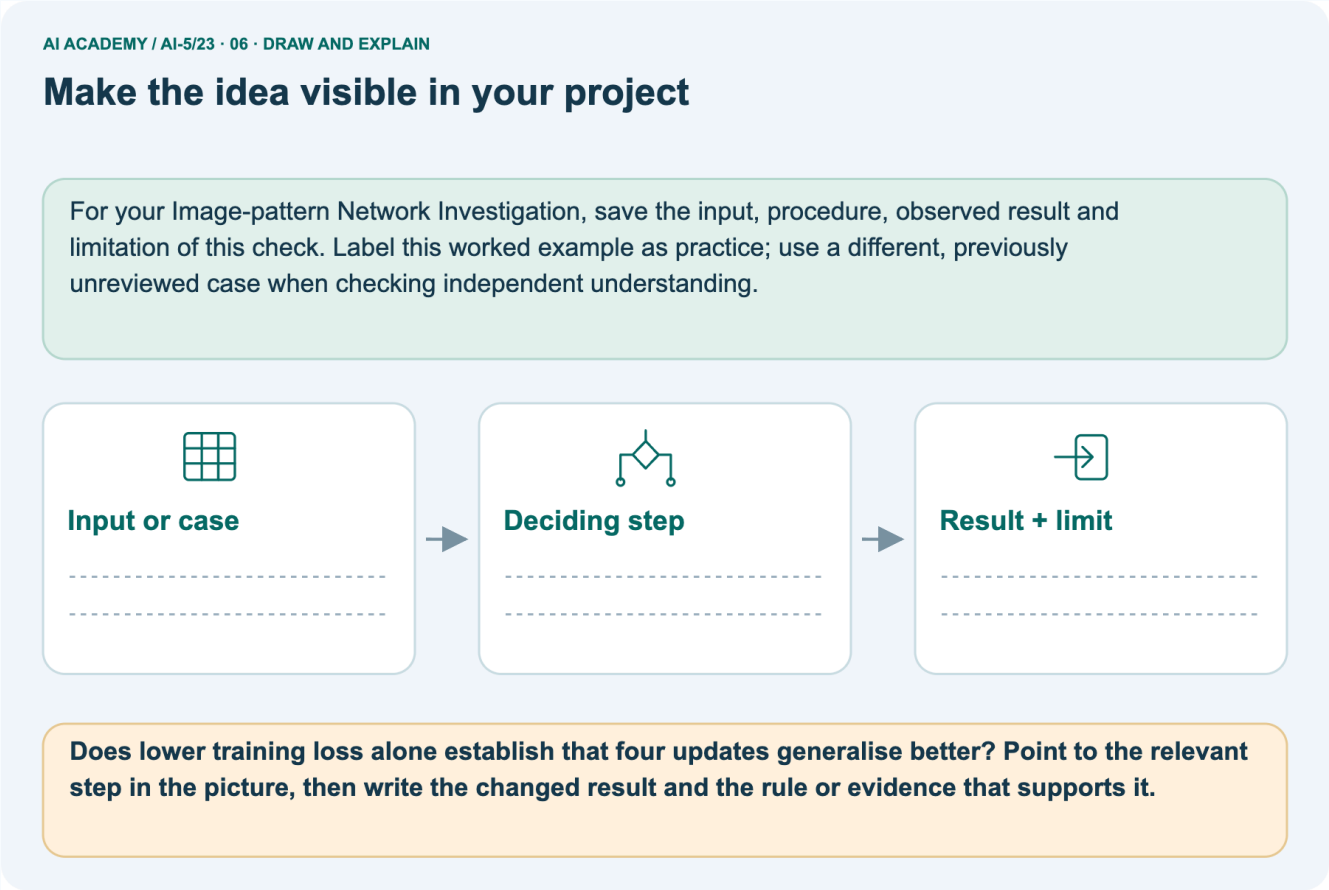
In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Training a small image model: the case

02. Follow the deciding steps

03. Compare the saved checkpoints

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.